

```
class HebbianResBlock(nn.Module):
    def __init__(self, dim):
        super().__init__()
        self.conv1 = HebbianConv2d(dim, dim, 3,
padding=1)
        self.conv2 = HebbianConv2d(dim, dim, 3,
padding=1)
        self.norm = nn.BatchNorm2d(dim)

    def forward(self, x):
        residual = x
        out = F.relu(self.conv1(x))
        out = self.norm(self.conv2(out))
        return F.relu(out + residual)

    def update_all(self, x, lr=0.01):
        y1 = self.conv1(x)
        self.conv1.local_hebbian_update(x, y1, lr)
        y2 = self.conv2(y1)
        self.conv2.local_hebbian_update(y1, y2,
lr)
```